

ISHARA HEWAWITHARANA

Computer Science Undergraduate

Phone: +94 7692088

Email: hewawitharanaishara@gmail.com

Address: No.1/4/A 1, Wela ihala, Handapangoda

LinkedIn: [linkedin.com/in/ishara-hewawitharana-701539333](https://www.linkedin.com/in/ishara-hewawitharana-701539333)

GitHub: github.com/ishara723



PROFESSIONAL PROFILE

Enthusiastic undergraduate Computer Science student with a strong foundation in programming, web development, and database management through academic coursework and practical projects. Focused on building backend applications using Spring Boot and working with MySQL and MongoDB databases. Seeking a software engineering internship to apply technical skills, gain hands-on industry experience, and contribute to real-world software development projects while learning from experienced professionals.

PROJECTS

Automated Waste Disposal System

[JAN 2024 - JUNE 2024]

Technologies and Tools: Raspberry Pi 3, OpenCV, Python, Servo Motors, Computer Vision Algorithms

- Developed an AI-driven automated waste segregation system utilizing computer vision to improve recycling efficiency.
- Integrated a Raspberry Pi with a camera module to capture real-time images and classify waste into categories like plastic, metal, and glass.
- Programmed motorized lid control using servo motors for accurate, contactless disposal based on object identification.

CIRO Mobile Phone Management System

[JAN 2025 - JUNE 2025]

Technologies and Tools: Spring Boot, MySQL, React, REST APIs

- Developed a full-stack mobile phone management system using Spring Boot, MySQL, and React.
- Implemented REST APIs and administrative CRUD operations for efficient management of phone records.

Distributed File Storage System

[JULY 2025 - NOV 2025]

Technologies and Tools: Python, Raft Consensus Algorithm, TCP Sockets, Vector Clocks, NTP

- Developed a fault-tolerant distributed storage system using the Raft consensus algorithm for leader election and log replication.
- Implemented real-time, heartbeat-based failure detection and a three-tiered consistency model (ONE, QUORUM, ALL) to ensure 99.2% availability.
- Utilized vector clocks for conflict resolution and TCP socket communication for authentic network connections between distributed storage nodes.

TestLang++: A DSL Compiler for Automated HTTP API Testing

[JULY 2025 - NOV 2025]

Technologies and Tools: Java 11+, JUnit 5, JFlex 1.9.1, CUP 11b, Java HttpClient, Spring Boot

- Designed a Domain-Specific Language (DSL) and compiler that translates .test scripts into executable JUnit 5 test classes.
- Automated API testing workflows by supporting GET, POST, PUT, and DELETE methods with built-in assertions for status codes, headers, and body content.
- Streamlined test maintenance through features like variable substitution and centralized base URL configuration.

Sarasavi Library Management System

[JULY 2025 - NOV 2025]

Technologies and Tools: Java, Spring Boot, MongoDB, React.js

- Developed an automated digital solution to streamline library operations, including real-time tracking of books, members, and reservations.
- Implemented secure role-based access control (RBAC) for students and administrators using Spring Boot backend services.
- Built a responsive frontend interface with React.js and integrated it with a scalable MongoDB database.

Coordina – Community Collaboration & Resource Platform

[JAN 2026 – MAY 2026]

Technologies and Tools: React.js, ASP.NET Web API, ADO.NET, Supabase, Docker, Azure, GitHub Actions, SignalR

- Developed a full-stack community collaboration and resource management platform with role-based access for events, donation drives, bookings, tasks, and communication.

- Implemented synchronized multi-level calendar systems with real-time updates using SignalR and integrated booking, chat, and task management features.
- Built secure backend APIs with ASP.NET and Supabase, utilizing Docker and Azure deployment with CI/CD pipelines and comprehensive testing workflows.

Tropical Reserve Rescue (AI Wildlife Rescue Simulation)

[JAN 2026 – MAY 2026]

Technologies and Tools: Unity, C#, Blender, A* Pathfinding, BFS, Raycasting, Rigidbody Physics, LineRenderer

- Integrated physics-based obstacle removal, smooth drone animation, and visual debugging using Unity tools such as Rigidbody, Raycasting, and LineRenderer.
- Implemented A* pathfinding, BFS fallback navigation, and graph-based node systems for autonomous rescue drone navigation and real-time path recalculation.
- Developed an AI-powered 3D wildlife rescue simulation in Unity set in a tropical eco-tourism reserve with dynamic environmental interactions.

EDUCATION

Bachelor of Science (Hons) in Computer Science

2024 - 2027

Sri Lanka Institute of Information Technology (SLIIT)

CERTIFICATIONS

- **2023:** Computing For Career Development – University of Colombo School of Computing (UCSC)
- **2020:** Web Designing Course – Zonal Computer Resource Centre, Taxila Central College

SKILLS

Technical Skills

Programming

Java, Python, C, JavaScript, SQL, C#

Web Development

HTML, CSS, React, Node.js, Spring Boot

Database Management

MySQL, MongoDB

Version Control

GitHub

Methodologies

Agile, Scrum

Debugging & Testing

Postman, Selenium, Junit, JMeter

Soft Skills

- Communication
 - Teamwork
 - Time Management
 - Leadership
 - Problem Solving
 - Adaptability & Learning
-

LANGUAGES

- English
- Sinhala